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EX NO: 8

Steering Angle Variation in MIMO Beamforming Using MATLAB

Aim

To analyze the effect of steering angle variation in a MIMO beamforming system by studying

steering angle, array gain, and beam direction error using MATLAB.

Software Required

- MATLAB

MATLAB Code:

```
clc;
```

```
clear;
```

```
close all;
```

```
angle = [-60 -30 0 30 60]; % Steering Angles (degrees)
```

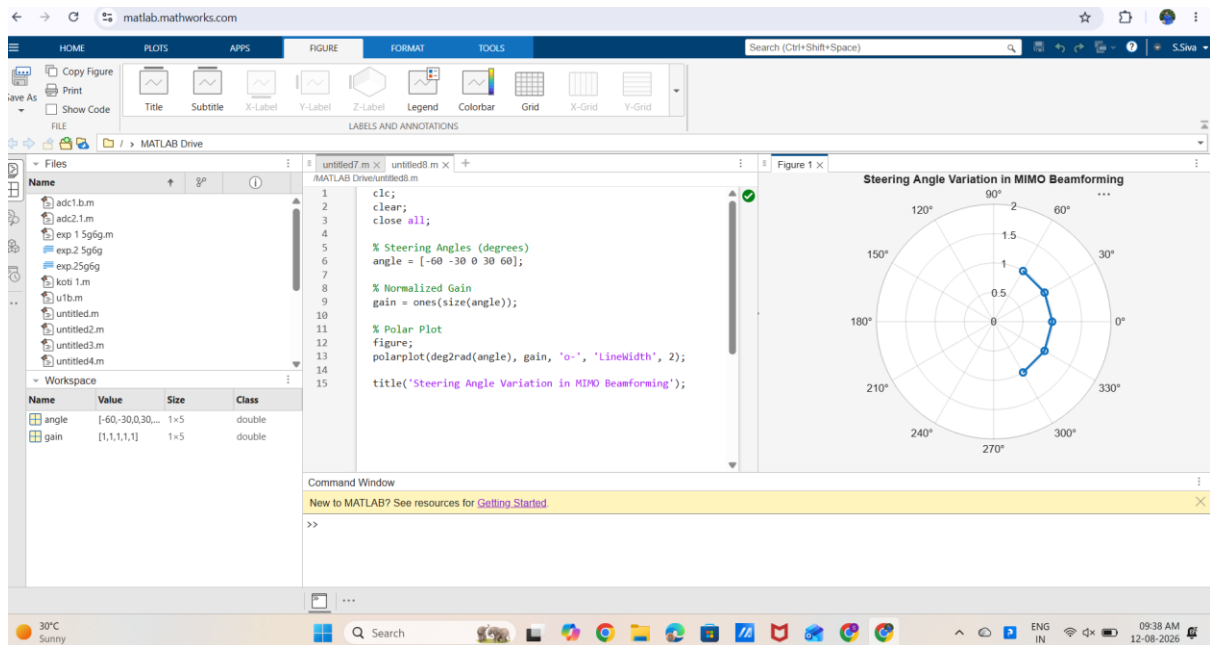
```
gain = ones(size(angle)); % Normalized Gain
```

```
figure
```

```
polarplot(deg2rad(angle), gain,'o-','LineWidth',2)
```

```
title('Steering Angle Variation in MIMO Beamforming')
```

OUTPUT:



Result:

- Plot showing different steering angles.
- Beam steering visualization.